

Range-wide Wood Thrush Motus Project Protocol

Permitting | Capture | Tags and Attachment | Feather Collection | Etc.

Prepared by Sarah Kendrick (USFWS; Sarah_Kendrick@fws.gov) and Nick Bayly (SELVA)



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Project Summary

The Wood Thrush (WOTH) is a priority species in 25 U.S. states and Canada. Improving our understanding of the species' full life cycle ecology, including migration patterns between their breeding and nonbreeding ranges, is essential in order to better understand the species' conservation needs throughout its range and to improve the design of targeted habitat management actions. With the above in mind, an international group of bird-conservation partners are initiating a hemispheric research and conservation project across the Wood Thrush breeding and nonbreeding ranges with the goal of deploying over 600 Motus tags, to better understand migratory connections, routes, timing, and survival across their full life cycle.

Project Permitting

Bird Banding Lab/Office Permit Authorization for Project

Project participants tagging Wood Thrush will need to get authorizations for Wood Thrush banding, auxiliary marker, and feather collection (1 feather per bird, 10 feathers per state) on U.S. BBL/Canadian BBO permits. *Before you apply*, the BBL has asked Sarah Kendrick to be the point of contact for these authorizations to make sure that all groups tagging WOTH are officially part of this common project. Therefore, please email Sarah at sarah_kendrick@fws.gov **before** you apply with the name, contact info, BBL/BBO permit number, and brief description of tagging experience of the lead bander, so this information can be relayed to the BBL. For those who have already sent this info to Sarah, please wait until this winter after the list has been shared with the BBL/BBO before applying for permit authorization.

Wildlife Collector/Collection Permits: State and Possibly Federal

Cooperators are also responsible for applying for your state/provincial scientific collection permit to trap, band, and tag Wood Thrush, where applicable. Also, if you are trapping and tagging Wood Thrush on federal lands, contact the area manager or biologist to inquire about additional permits or permissions needed. Note that you may need BBL/BBO authorization prior to applying for these permits.

Institutional Animal Care and Use Committee (IACUC) Approval

FWS Migratory Bird staff are exempt from IACUC approval and there is no IACUC requirement at this federal level to acquire blanket IACUC/ACUC approval for the whole project. However, the common project protocol outlined here has been approved by USFWS leadership in Region 3 and the USGS Bird Banding Lab; letters of support for the project and its protocol are included within this document. Since the protocol development has occurred under the purview of the FWS (Sarah Kendrick) and the project and protocol are approved by the BBL, this support should assist you in your IACUC/ACUC applications, should you need them. If you require IACUC approval at your institution or agency, please apply several months before your field work to allow for processing time.

Capture

All Wood Thrush tagged for this study should be captured and tagged on breeding territories within range of an active Motus station to allow for local tracking prior to migration.

We recognize that with such a wide range of target-netting experience and a wide range of Motus station coverage by state, it is difficult to set stringent rules for which type of WOTH individuals that

should be trapped and tagged. Therefore, we suggest a few different strategies based on your experience:

1. WOTH-tagging Beginner: Catch any territorial adults you can (males likely easier to catch).
2. Nest Method: Find nests and net an adult near the nest.
3. WOTH-tagging Pro: Try to deploy half of your tags on adults and half on juveniles.

Do not deploy tags on nestlings for this project.

When do I tag WOTH?

Please target-net and tag birds on breeding territories. We do not want to tag passage birds. A good way to select a start date for trapping and tagging is to use the BBS-survey start date where you will be tagging, when birds are assumed to be on territories and done migrating (in Missouri, for example, we will trap starting the last week in May through early to mid-June). Do not trap and tag WOTH past mid-August when they have started to move around and migrate. Trapping later in the breeding season (July, early August) will also be much more difficult, so do not wait until July to tag birds and risk not deploying all of your state's tags. Optimum dates for deployments across most of the eastern U.S. will likely be end of May through mid-June.

Which WOTH do I target for tagging?

Best case scenario: Deploy a Motus-tag on one individual WOTH per territory/family group to better track birds at a population level and spread out the tags. If you have the option and there several suitable WOTH sites with Motus stations in your state, aim to tag a minimum of 5 WOTH within range of any one Motus station (as opposed to the extreme example of tagging 1-2 WOTHs at many sites).

If you do not have enough Motus stations near suitable WOTH habitat and you aren't able to tag just one individual per family group, deploy the 25 tags on any WOTH you are able to catch within range of an active Motus station.

Weight minimum: do not tag any bird under 45g (unless you have been granted approval for 4% or 5% body weight threshold by the BBL on your permit). 45g is our minimum weight threshold for Lotek tags to not exceed 3% of the bird's body weight, as per federal regulations. However, you can request BBL grant you an increased 4% or 5% body weight for auxiliary markers on your permit.

Mist Netting and Conspecific Playback

Project partners are not required to adhere to the same trapping techniques across the project, but below are some tips and tricks for target netting WOTH from researchers who have captured many. If you find another approach to target-net birds that works for you, great. Here we provide some ideas and approaches:

- Wood Thrush can be target-netted using mist nets and conspecific playback (a mix of songs and *bupbupbup* calls). A mix of these sounds can be found and downloaded at <https://xeno-canto.org/species/Hylocichla-mustelina>.
- Set up 1-3 nets connected in a V, Y, or T shape with the playback source placed in the center. Nets with a mesh size of 38 mm will reduce the chances of birds bouncing or escaping, but 30mm mesh will work.

- If possible, set up net arrays in shrubby areas so they are better concealed. Another approach would be to set up multiple net arrays but this can be more time consuming than a more targeted approach.
- Set up multiple speakers within a breeding territory with playback loop uploaded if possible (not connected to phone by Bluetooth) and distance yourself from the nets or leave the immediate netting area so birds will not be spooked by your presence. Check nets from a distance every 10-20 minutes.
- If you can spare the time, take a few minutes to silently follow birds at a distance to find their nest. Watch the male and female approach the nest a few times to learn their routes/perches used to get to the nest and set up nets here. This is most effective if the nest is below 3-4m. Do not attempt to net WOTH at a nest for longer than 30 minutes at a time.
- For playback, a mix of song recordings and alarm calls/*bupbupbup* varying in intensity and repeated on a long loop works best.
- If you have access to a WOTH decoy (stuffed real bird is preferable), use it. For some, birds have responded better when the decoy is in the middle of a funneled net array half-way up the net.

Special thanks to Callie Stanley, Brendan Boyd, Rémi Torrenta, Nick Bayly, and others for their insights.

Tags: Specs, Harnesses, and Attachment

Project partners should deploy *Lotek nanotags* with an identical setup for this range-wide project to allow for a single analysis of the data from all tags deployed (potential of 600+ tags), without having to account for difference in tag type or setup. This approach will maximize the impact and use of project findings. A subset of CTT hybrid tags will be deployed by the project's core team to test an alternate tag method in areas where Lotek tags are also being deployed. Lotek tags are the best fit for this study given the current Motus receiving station network, as more stations are capable of detecting the 166.380 frequency than other tag types in our geography of interest, especially in the Neotropics where we need as many tag detections as possible. Lotek tag specifications for the project can be found below.

Lotek Tag Purchase Options

Partners can purchase their own Lotek tags but please register them (at purchase) with Lotek under your state's Motus Project number. Email Sarah_Kendrick@fws.gov to get this number. Alternatively, partners can pool funds for a large group tag order that will be placed in late January/February. Lotek tags cost \$245 each (\$225 per tag + \$20 Motus fee per tag). Funds can be pooled at the Missouri Conservation Heritage Foundation (MCHF) online by credit card at <https://mochf.org/wood-thrush-motus-tagging-project/> or by sending a check to Missouri Conservation Heritage Foundation; P.O. BOX 366; Jefferson City, MO 65102 – 0366 – please note on any online contributions or checks that the funds are for the *Wood Thrush Motus-tagging Project*. The tag order from pooled MCHF account funds will be placed by February 1, 2024 and February 26, 2025, respectively; please get funds into the account prior to that time if you are not purchasing your state's tags.

If partners purchase their own tags, inform Lotek you are ordering tags for your state's Motus Project (will be named WOTHParty_[your 2-letter state abbreviation]). They will know the tag specs you need. However, they are listed here:

Nanotag Model: NTQB2-4-2-M
HD potting and antenna
15 cm antenna
2 x 1.5mm inner diameter tubes
14.9s or 15.1 burst interval

Lotek Contacts

There are two contacts at Lotek for tag orders as part of this project depending on where you are located. If you are ordering your state's tags yourself, please contact the appropriate Lotek salesperson:

- Alan Marsh (amarsh@lotek.com) if you are ordering tags for MO, IA, MN, AR, LA and anywhere west or south through Mexico to South America.
- Mike Van den Tillaart (mtillaart@lotek.com) if you are ordering tags in WI, IL, KY, TN, MS and anywhere further east.

Tag Harness

Do not tag any WOTH that weighs less than 45g (unless you have been granted approval for 4% or 5% body weight threshold by the BBL on your permit). This is our weight minimum for tags to not exceed 3% of the bird's body weight. Tags and harness materials weigh 1.5g.

All tags must be fitted to WOTH using the same harness type and materials across the project to ensure that tags remain on the bird for over a year. We appreciate all partners using these same methods. This design allows you to build and attach the harnesses to the tags before going into the field, as the harness only needs to be tightened and crimped after achieving the appropriate fit on the bird.

Tagging materials:

- 0.7 mm Stretch Magic Bead Cord (clear or black, but black is easier to see)
- Beadalon crimp, Size #2
- Crimp pliers or other small pliers
- Small scissors
- If you do not have a Lotek Tag Activator, you will need to purchase one to active your Motus tags prior to deployment; the cost is \$158.

Harness Building and Attachment: Descriptions and Videos

Harnesses will consist of a figure-8 leg-loop harness using 0.7 mm Stretch Magic and one Beadalon Size #2 crimp. The flat side of the tag should sit against the bird's back as close to the middle as possible; the numbers on the tag should be pointing away from the bird's back. Cut a piece of Stretch Magic about 24 cm long and feed through the top loop of the tag. Take that end of the Stretch Magic and go back through the tube to make a loop around the top of the tag centered on your length of Stretch Magic (Figs. 1A and 1B). Feed both loose ends of the Stretch Magic through the bottom tag tube to create your leg loops (Fig. 2). Feed both loose ends of your Stretch Magic through the crimp in the same direction (down, toward the antenna; Figs. 3A and 3B).

Fig. 1A

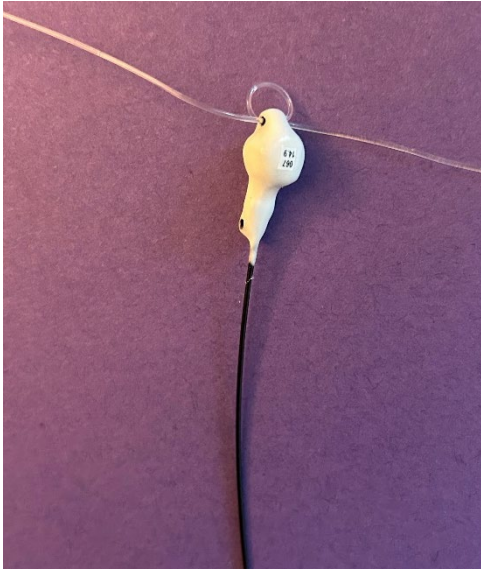


Fig. 1B

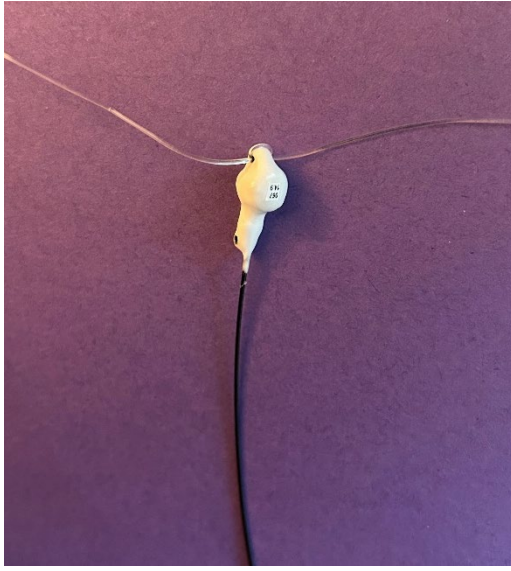


Fig. 2

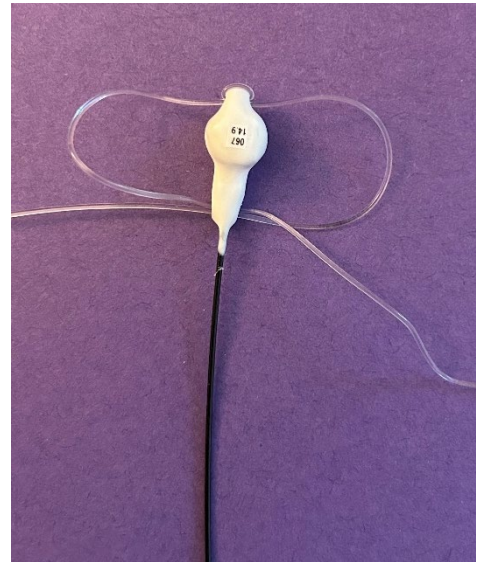


Fig. 3A

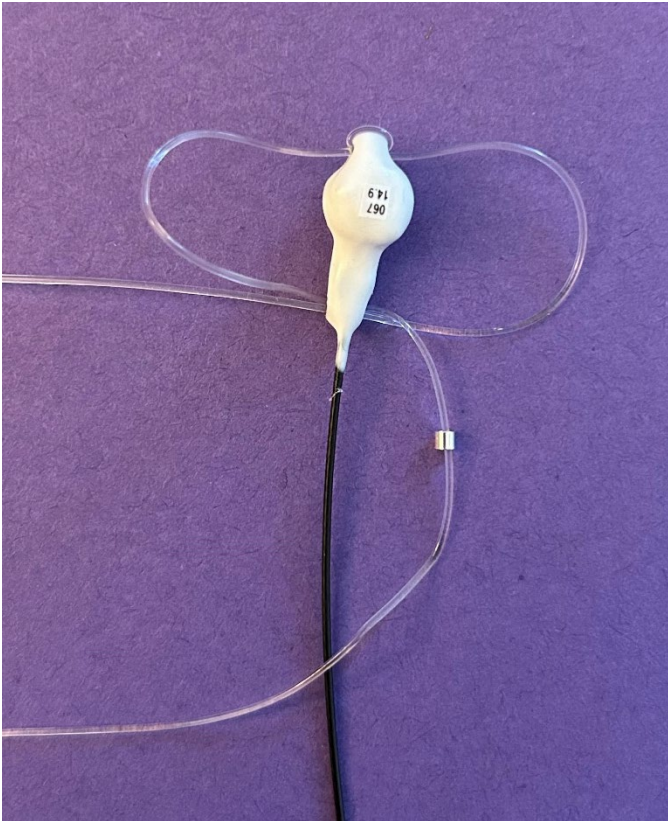
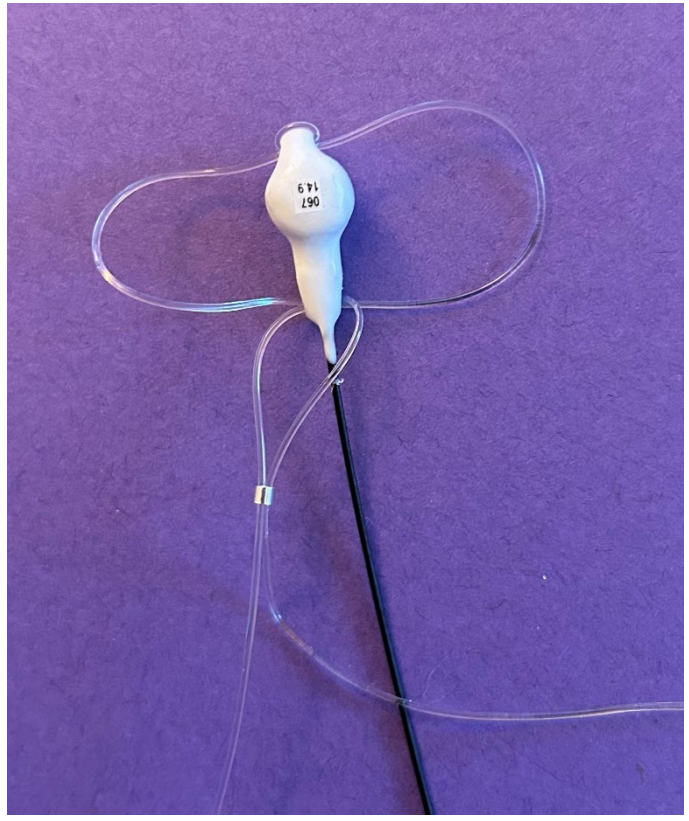


Fig. 3B



DON'T FORGET TO ACTIVATE YOUR MOTUS TAGS PRIOR TO PLACING THEM ON BIRDS.

Either activate tags the morning of trapping or activate the tag immediately prior to placing on the bird.

To fit a tag to a bird, slide one leg loop up to the top of the thigh on one leg, and then wrap the tag and harness over the back of the bird and fit the second leg loop to the opposite leg. Once the tag is in position, high on the thighs of both legs, adjust the leg loops to the bird. While one person holds the bird with the leg loops fitted properly, have a second person slide the crimp up to the tag so the leg loops will not loosen. Clamp the crimp using crimping pliers or banding pliers (Fig. 5) and cut excess the stretch magic. Voila (Fig. 6 completed tag without a bird)!

Fig. 4

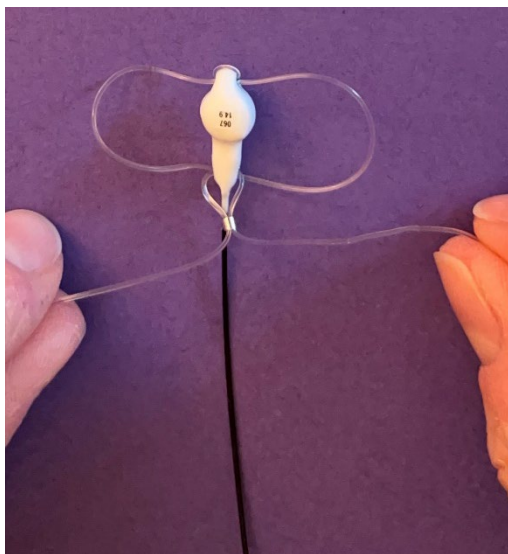
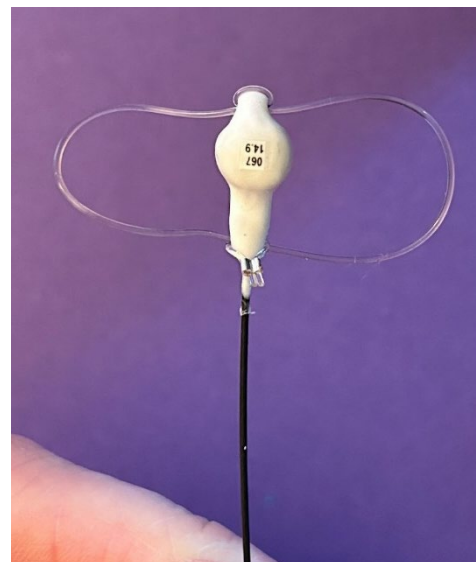


Fig. 5



Fig. 6



Carefully check tag fit, making sure that the bird can freely move both its legs and wings (flap), and that the harness is not over the top of any of the flight feathers e.g., tertials. The leg loops should be visible at the top of the bird's thighs against its body and **not** halfway down the thigh. The ideal tag fit should allow for the tip of your finger to be partly inserted under the tag but not so loose that the tag can be "flipped over". Another way of checking for a loose fit, is to gently pull down on the tag while holding the antenna. If the tag slips off the thighs it is too loose.

Once tag fit has been carefully assessed and ensuring that the harness is well crimped, release the bird low to the ground. If the bird does not fly off normally, cut off the tag and try again; the crimped crimp will not allow for easy resizing.

For fitting the harnesses and assessing fit, we strongly recommend that all cooperators watch the video below. *Note: To be authorized by the Bird Banding Lab for WOTH auxiliary markers (tags), previous experience tagging birds or hands-on tagging training is necessary. The how-to video below does not take the place of or serve as training, but rather as a guide for attaching this harness type to a thrush.*

How-to videos for harness building, attachment, and assessing fit:

- Harness building video [available here](#).
- Harness attachment video [available here](#) on a Gray-cheeked Thrush. This video shows a tag with 2 crimps at the top of the tag. Rather than this approach, we recommend the Stretch Magic looped through the top tube (no crimps) as described above and in the *Harness building video*.

Feather Collection

To participate in feather collection, you must include feather collection on your BBL application and be granted authorization.

Please collect the innermost primary covert from 10 birds tagged in your state. Place these feathers into individual paper envelopes with the following information written on the envelope: species, date, state, lat/long or UTM of the capture site, band number, and tag ID. The innermost primary covert (PPcov 1) is shown in the image below. These feathers will assist in isotopic signature validation across the species' range as part of a separate analysis. We do not need more than 10 samples per state. Please store in a cool, dry place and ship to Sarah Kendrick after each field season: 101 Park De Ville Dr. Columbia, MO 65203.



Fig. 7: Single feather to be collected on no more than 10 birds/state or province: innermost primary covert (PPcov 1).

Photo of a Swainson's Thrush by Nick Bayly

Optimizing Motus Stations at Tagging Sites

If possible, we encourage participants to optimize your Motus stations to maximize the detections of locally tagged Wood Thrush. Add antennas that cover station blind spots or the direct where most WOTH were tagged. This part of the project is not mandatory but can improve local tracking of birds and also increases the chances of capturing a southerly migratory movement post-breeding if you have the resources to do so. Adding an antenna to your station will require an additional FunCube radio for each antenna added; they can be purchased at the [FunCube website \(UK\)](#). Radios are cheaper from FunCube directly, but you'll have to process them by installing the correct firmware. If you can't purchase outside

the U.S., you can also buy FunCubes from [RFS Scientific](#) with the firmware already downloaded and ready to use. Even if the website indicates FunCubes are out of stock, email Rhonda (rfsscscientific@gmail.com) to check.

Collaborator Policy

Project collaborators and those deploying tags will be clearly acknowledged in future manuscripts resulting from project-wide analyses of these data. Co-authorship is an option for those who contribute directly to the analysis and manuscript writing. Please contact Sarah (sarah_kendrick@fws.gov) if interested, but be aware that data will be collected (with live Motus tags on the landscape) until roughly winter 2026-27, so analyses will not be conducted until this time, with manuscript writing to follow.

Example Single Feasible Source Paperwork Language for State Agencies

If helpful, state agencies can use this blurb for Single Feasible Source Paperwork if you are purchasing your own Lotek tags. Lotek is an international vendor (Canada), so additional paperwork may be necessary for state agencies to make this purchase.

The Motus Wildlife Tracking System (Motus) is a collaborative research network that uses arrays of automated radio telemetry receivers to study movements of small animals. Motus works using ultra lightweight radio tags attached to our smallest migratory birds and bats that are coded to the Motus frequency. When these Motus-tagged animals fly past a receiver on the landscape, the signal of the tag is detected and stored. Lotek Wireless Inc. exclusively manufactures Motus-digitized radio tags on the 166 MHz frequency. No other company manufactures these tags.

WOTH Processing and Tagging Checklist

- ☐ Band the bird (size 1A)
- ☐ Activate and attach tag and check fit: proper leg movement and wing flap
- ☐ Age, Sex
- ☐ Weight
- ☐ **Not necessary in 2025:** Feather sample – PPcov #1, up to 10 collected per state
- ☐ Record all information on Project Datasheet below!
Note that this Project Datasheet does not take the place of regular submission of banding records to BBL.
- ☐ At end of season, submit all data to Sarah_Kendrick@fws.gov or submit in online format as per Sarah's instructions.

Sexing and Ageing Guidelines

Sexing: For the purposes of this project, we will only sex birds on the breeding grounds if they have a cloacal protuberance (male) or brood patch (female). Be aware that brood patch regression has been seen as early as 25 June in Delaware, so if you catch birds later in the season still on breeding territory, sexing by BP/CP may be indeterminate. If so, just record as such on the datasheet.

Ageing: Record age as Adult (ASY), Second Year (SY) or Juvenile (J). If you target-net early in the season, Juvenile will not be an option. If netting after a full nest attempt could have occurred, age Juveniles using the following criteria from Pyle (2nd ed.):

Adult (ASY or DCB: May-July): Prior to the full annual molt, adult birds should have uniform plumage with no buffy tips or contrasts within the greater coverts or between the greater coverts and the median coverts.

Second year (SY or FCF: May-July): Birds in their first breeding season will still maintain retained outer greater coverts with buffy tips and have pointed tail feathers, until undergoing their post-breeding molt. Presence of a brood patch or cloacal protuberance, as well as worn flight and tail feathers, will clearly differentiate these birds from birds born in the current breeding season.

Juvenile (J or FCJ: May-Aug): Juvenile birds have buffy spots or tips to the feathers of the upperparts (head and back), as well as in the lesser, median and greater coverts; also note fleshy gape in recently fledged birds. See Figure 5 below.

Adult birds caught later in the season will likely be undergoing a complete molt and should be aged as DPB or SPB using the WRP system, while most birds hatched in the same season will have initiated or completed their preformative (post-juvenile) molt and should be aged as FPF or FCF respectively. Ideally, birds that have completed their preformative molt and are aged as FCF in July or August, should not be tagged, as they may originate from other breeding populations. They can be identified as having retained outer greater coverts that are shorter, slightly paler, and generally with small buffy tips, which contrast with replaced inner greater and median coverts.



Fig. 5: Juvenile (FCJ) Wood Thrush
Photo by Bracken Brown, Willistown
Conservation Trust

Letters of Support for Project and Protocol

Support letters below from US Fish and Wildlife Service Assistant Regional Director Brian Smith and the USGS Bird Banding Lab's Program Manager Tony Celis. Higher-res PDFs of these documents are provided as attachments in the Protocol email or by request from Sarah_Kendrick@fws.gov.



United States Department of the Interior

FISH AND WILDLIFE SERVICE

5600 American Boulevard West, Suite 990
Bloomington, Minnesota 55437-1458



November 21, 2023

To Whom It May Concern:

This letter serves as approval and support of the Wood Thrush Motus-tagging Project protocol. This project will deploy over 600 Motus tags on Wood Thrush in over 15 states and one Canadian province across the breeding range and work with international partners to deploy tags across the wintering range in five countries. The project goals are to better understand migratory connections, routes, timing, and survival across the full life cycle to inform future conservation and management for this Species of Greatest Conservation Need in 25 states across its breeding range.

On behalf of the U.S. Fish and Wildlife Service Region 3 Migratory Bird Program, we fully endorse and support this project led by Sarah Kendrick of our regional program. The U.S. Geological Survey Bird Banding Lab has reviewed and approved the project proposal for use by her and other cooperators. We thank you in advance for your support of this project.

Sincerely,

BRIAN SMITH Digitally signed by BRIAN SMITH
Date: 2023.11.27 11:47:13 -05'00'

Brian Smith, Ph.D.
U.S. Fish and Wildlife Service, Midwest Region
Assistant Regional Director, Migratory Bird Program
2651 Coolidge Rd, Suite 101
East Lansing, MI

Phone: 303-656-6117
Email: brian_w_smith@fws.gov

Support letter from USGS Bird Banding Lab. Higher-res PDF of this document is provided as an attachment in the Protocol email.



United States Department of the Interior

U.S. GEOLOGICAL SURVEY

Eastern Ecological Science Center
12100 Beech Forest Road
Laurel, Maryland 20708
(301) 497-5500

December 4, 2023

To Whom It May Concern,

Please find this letter in support of the U.S. Fish and Wildlife Service's collaborative project: *Range-wide Motus Tracking of Wood Thrush to Inform Conservation across the Full Annual Cycle*. This project, managed by USFWS Biologist Sarah Kendrick, aims to track the movements of Wood Thrush throughout their range using automated radio telemetry. This technique utilizes a tracking method where digitally encoded radio transmitters (aka Nanotags) are deployed on individuals that allow them to be detected by the MOTUS network of receiving towers during their migration. Transmitters will be deployed in 15 states on the breeding grounds, as well as numerous sites on the wintering grounds, to gain insights into both spring and fall migration patterns.

As the federal permitting body for banding and marking in the United States, the U.S. Geological Survey's Bird Banding Laboratory has reviewed the initial project proposal and approves of the overall banding and marking plans within. However, this support does not take the place of official review of permit requests or applications submitted by individual collaborators. Because federal banding and marking permits are issued at the local state or county level, collaborators must still apply for a federal banding and marking permit or request a modification to their existing permit including a detailed description of their local banding methods and documentation of the knowledge and skills necessary to do the proposed work.

Sincerely,

ANTONIO

CELIS MURILLO

Digitally signed by ANTONIO
CELIS MURILLO
Date: 2023.12.05 16:04:02
-05'00'

Antonio Celis-Murillo, PhD
U.S. Geological Survey
Eastern Ecological Science Center
Program Manager, Bird Banding Laboratory (BBL)
12100 Beech Forest Rd. Laurel, MD 20708
Office: 301-497-5808

Wood Thrush Motus-tagging Project Datasheet

Sex: Age by CP/BP. If indeterminate, enter U (Unknown).
Age: Adult (**A**), Second Year (**SY**), or Juvenile (**J**). Adults will have a uniform wing. SY have buffy tips on the outer greater coverts and Juveniles will have buff spots on feathers and fleshy gape. If Juvenile, note whether FCJ or FCF (see Sexing and Ageing Guidelines in Protocol).

Bird No.	State	Site Name	Location: UTM or Lat/Long	Date	Tagger Initials	Capture Time	Species	Band Size	Band No.	Tag ID (activate!)	Age	Sex	How Sexed CP/BP	Body Mass (g)	PPCov1 Collect? Y/N	Notes
1							WOTH	1A	-							
2							WOTH	1A	-							
3							WOTH	1A	-							
4							WOTH	1A	-							
5							WOTH	1A	-							
6							WOTH	1A	-							
7							WOTH	1A	-							
8							WOTH	1A	-							
9							WOTH	1A	-							
10							WOTH	1A	-							
11							WOTH	1A	-							
12							WOTH	1A	-							
13							WOTH	1A	-							
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16							WOTH	1A	-							
17							WOTH	1A	-							
18							WOTH	1A	-							
19							WOTH	1A	-							
20							WOTH	1A	-							
21							WOTH	1A	-							
22							WOTH	1A	-							
23							WOTH	1A	-							
24							WOTH	1A	-							
25							WOTH	1A	-							

This content copied/pasted from above in case you want to print datasheet front and back with this reference material:

WOTH Processing and Tagging Checklist

- ☐ Band the bird (size 1A)
- ☐ Activate and attach tag and check fit: proper leg movement and wing flap
- ☐ Age, Sex
- ☐ Weight
- ☐ NA for most partners in 2025: Feather sample – PPcov #1, up to 10 collected per state
- ☐ Record all information on Project Datasheet below!
Note that this Project Datasheet does not take the place of regular submission of banding records to BBL.
- ☐ At end of season, submit all data to Sarah_Kendrick@fws.gov or submit in online format as per Sarah's instructions.

Sexing and Ageing Guidelines

Sexing: For the purposes of this project, we will only sex birds on the breeding grounds if they have a cloacal protuberance (male) or brood patch (female). Be aware that brood patch regression has been seen as early as 25 June in Delaware, so if you catch birds later in the season still on breeding territory, sexing by BP/CP may be indeterminate. If so, just record as such on the datasheet.

Ageing: Record age as Adult (ASY), Second Year (SY) or Juvenile (J). If you target-net early in the season, Juvenile will not be an option. If netting after a full nest attempt could have occurred, age Juveniles using the following criteria from Pyle (2nd ed.):

Adult (ASY or DCB: May-July): Prior to the full annual molt, adult birds should have uniform plumage with no buffy tips or contrasts within the greater coverts or between the greater coverts and the median coverts.

Second year (SY or FCF: May-July): Birds in their first breeding season will still maintain retained outer greater coverts with buffy tips and have pointed tail feathers, until undergoing their post-breeding molt. Presence of a brood patch or cloacal protuberance, as well as worn flight and tail feathers, will clearly differentiate these birds from birds born in the current breeding season.

Juvenile (J or FCJ: May-Aug): Juvenile birds have buffy spots or tips to the feathers of the upperparts (head and back), as well as in the lesser, median and greater coverts; also note fleshy gape in recently fledged birds. See Figure 5 below.

Adult birds caught later in the season will likely be undergoing a complete molt and should be aged as DPB or SPB using the WRP system, while most birds hatched in the same season will have initiated or completed their preformative (post-juvenile) molt and should be aged as FPF or FCF respectively. Ideally, birds that have completed their preformative molt and are aged as FCF in July or August, should not be tagged, as they may originate from other breeding populations. They can be identified as having retained outer greater coverts that are shorter, slightly paler, and generally with small buffy tips, which contrast with replaced inner greater and median coverts.

